

PROCESS DESIGN DOCUMENT

(PDD)

Blinkit Sales Performance Dashboard — Power BI Analytics Solution

Document Type	Process / Project Design Document
Related Artefact	blinkit_dashboard.pbix
Platform	Microsoft Power BI Desktop
Version	1.0
Status	Draft — Reverse-engineered from delivered .pbix file

1. Introduction & Purpose

This Process Design Document (PDD) describes the business context, objectives, scope and process design behind the **Blinkit Sales Performance Dashboard**, a Power BI report built to analyse retail/quick-commerce outlet sales performance. The document has been prepared by reviewing the structure of the delivered Power BI file (**blinkit_dashboard.pbix**), including its data fields, measures, report page, visuals and interactivity design.

The purpose of this PDD is to capture **what business problem the dashboard solves, who it is for, and what the end-to-end process looks like** from raw sales data to the insights presented on screen — independent of the technical implementation, which is covered separately in the accompanying System Design Document (SDD).

2. Business Background & Problem Statement

Quick-commerce and retail grocery businesses (such as Blinkit) operate across a large number of outlets that vary by size, location type, and store format. Item-level sales also vary widely by product category and packaging/fat-content classification. Without a consolidated view, decision-makers struggle to answer basic but important questions such as:

- Which item categories generate the highest sales, and which underperform?
- How does outlet size, location type, or outlet format (Grocery Store, Supermarket Type 1/2/3, etc.) influence sales and customer ratings?
- Are newer or older outlets (by establishment year) performing differently?
- How is overall sales performance trending, and how many outlets contribute to the total network?

The dashboard was built to consolidate this information into a single, interactive report so that business stakeholders can explore sales and rating performance without needing to query raw data manually.

3. Project Objectives

- Provide a single-page, at-a-glance summary of overall sales performance (Total Sales, Average Sales, Total Outlets, Average Rating).
- Enable breakdown and comparison of sales by Item Type, Item Fat Content, Outlet Size, Outlet Type, Outlet Location Type, and Outlet Establishment Year.
- Allow interactive filtering via slicers, so users can narrow the view to a specific outlet size or location type.
- Provide a detailed, cross-tabulated (pivot table) view for users who need granular figures behind the visual summary.
- Present the report with clear Blinkit branding and a clean, modern, dark-themed design suitable for stakeholder presentation.

4. Scope

In Scope

- Analysis of historical sales, ratings, and outlet attributes contained in the source dataset.
- A single interactive Power BI report page with KPI cards, charts, slicers, and a detailed pivot table view.
- Toggle-based navigation between a chart-based summary view ("HOME") and a detailed tabular view ("TABLE") using bookmarks.

Out of Scope

- Real-time or automated data refresh from a live transactional/POS system (the current file reflects a static, imported dataset).

- Row-Level Security (RLS) or user-specific data access restrictions — none were found configured in the file.
- Predictive analytics or forecasting (e.g. future sales prediction); the report is descriptive/diagnostic in nature.
- Multi-page drill-through reports — the solution is delivered as a single consolidated report page.

5. Stakeholders & Roles

Role	Responsibility
Business / Operations Stakeholder	Consumes the dashboard to review sales and outlet performance and make operational decisions.
Data / BI Analyst (Report Author)	Designed the data model, authored DAX measures, and built the report visuals and navigation.
End User	Interacts with slicers and navigation buttons to explore filtered views of the data.

6. Data Source Overview

The report is built on a single flat data table (referred to in the model as **Table1**), consistent with a typical retail outlet/item sales dataset (of the kind popularly used in Blinkit / BigMart-style sales-analysis exercises). Based on the fields actively used across the report's visuals, the dataset includes at minimum the following business attributes:

- Item Type — the product category of the item sold
- Item Fat Content — packaging classification (e.g. Low Fat / Regular)
- Outlet Size — Small / Medium / High
- Outlet Type — e.g. Grocery Store, Supermarket Type 1/2/3
- Outlet Location Type — Tier 1 / Tier 2 / Tier 3 style classification
- Outlet Establishment Year — the year each outlet was set up
- Underlying sales and rating figures used to derive the report's measures

Note: the Power BI data model is stored internally in a compressed, engine-level format, so the full underlying column list and row-level data could not be directly extracted for this document. The list above reflects only the fields actively referenced by the report's visuals. The complete column list can be confirmed by opening the .pbix file in Power BI Desktop.

7. End-to-End Process Flow

- **Step 1 — Data Ingestion:** Source sales/outlet data is loaded into Power BI (imported, not a live/DirectQuery connection).
- **Step 2 — Data Modelling:** The data is held as a single table (Table1), supplemented by a dedicated measures table ("DAX MEASURES") and a small helper table ("Metrics") used to drive dynamic selections in the report.
- **Step 3 — Calculation Layer:** Business metrics (Total Sales, Average Sales, Total Outlets, Average Rating) are defined as DAX measures on top of the raw data.
- **Step 4 — Report Design:** A single report page is designed with KPI cards, charts (funnel, donut, clustered bar, area) and a detailed pivot table, styled with Blinkit branding.
- **Step 5 — Interactivity:** Slicers (Outlet Location Type, Outlet Size, Metrics) allow the user to filter the report; two bookmark-linked buttons ("HOME" and "TABLE") let the user switch between the chart summary and the detailed table view.
- **Step 6 — Consumption:** End users open the report in Power BI Desktop/Service, filter as needed, and draw operational insights.

8. Functional Requirements

ID	Requirement
FR-01	The report shall display Total Sales, Average Sales, Total Outlets, and Average Rating as headline KPI cards.
FR-02	The report shall allow users to filter data by Outlet Location Type and Outlet Size via slicers.
FR-03	The report shall visualise sales by Item Type (funnel) and Item Fat Content (clustered bar chart).
FR-04	The report shall visualise sales and rating trends by Outlet Establishment Year (area chart).

ID	Requirement
FR-05	The report shall provide a donut chart comparing Average Rating and Average Sales by Outlet Size.
FR-06	The report shall provide a detailed pivot table cross-tabulating Item Type, Outlet Size and Establishment Year against Outlet Type.
FR-07	The report shall allow the user to switch between a chart-based summary view and the detailed table view via on-screen navigation buttons.

9. Non-Functional Requirements

- **Usability:** The report must be understandable at a glance, with a clear visual hierarchy (KPI cards at top, charts and table below).
- **Performance:** As the model uses a single imported table with no complex relationships, the report should render and filter with minimal latency.
- **Portability:** The report is delivered as a self-contained .pbix file that can be opened on any machine with Power BI Desktop.
- **Branding/Look & Feel:** The report must reflect Blinkit's brand identity (black and green "BLINKIT" wordmark) on a dark theme background.

10. Assumptions & Constraints

- The dataset represents a point-in-time snapshot of outlet and item sales; it is not connected to a live transactional source.
- No Row-Level Security (RLS) roles were found configured, so all users of the file see the same data.
- Exact DAX formulas behind each measure are stored in Power BI's internal compressed data model and were inferred from field usage and naming rather than extracted verbatim; they should be verified directly in Power BI Desktop.
- The report is designed for single-page consumption and assumes a 1920×1080 display/export canvas.

11. Deliverables

- blinkit_dashboard.pbix — the Power BI report file
- This Process Design Document (PDD)
- The accompanying System Design Document (SDD)

12. Success Criteria

- Stakeholders can identify top and bottom-performing item categories and outlet types within seconds of opening the report.
- Users can filter the report by outlet attributes without needing support from the report author.
- The detailed pivot table view provides sufficient granularity to answer follow-up questions raised by the summary charts.

13. Risks & Mitigations

Risk	Mitigation
Static dataset becomes outdated over time	Establish a periodic refresh/re-import process, or connect to a live source in a future iteration.
Single flat table limits scalability if new data dimensions are added	Consider migrating to a proper star schema (fact + dimension tables) if the model grows.
No RLS means all users see all data	Introduce RLS roles if the report is shared with audiences that should see restricted subsets.

Prepared based on structural analysis of the supplied .pbix file. Refer to the System Design Document (SDD) for technical/data-model details.